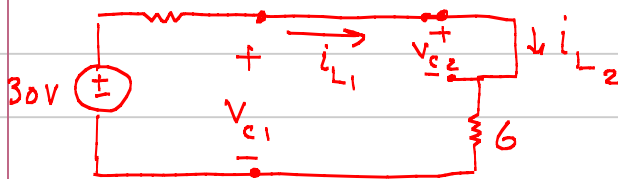




Ex. Find the capacitor voltages and inductor currents under dc condition. (Under dc condition all voltages and currents are constant).

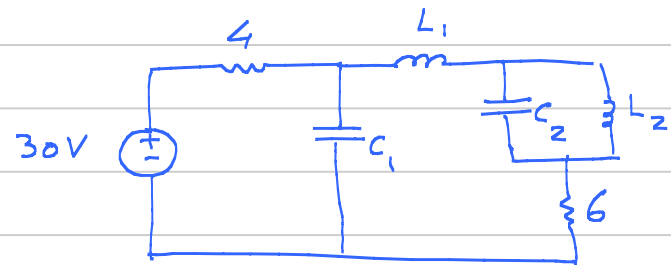
Under dc condition, capacitors are open and inductors are short circuit:



$$i_{L1} = i_{L2} = \frac{30}{4+6} = 3A$$

$$V_{C1} = 6 i_{L2} = 18V$$

$V_{C2} = 0$ (voltage across a short ckt. is zero)



Chapter 4, Transient

In this chapter we will focus on circuits containing sources, resistors, switches, inductors and capacitors.

RL and RC circuits are known as the first order circuits because their voltages and currents are described as first order differential equations.

RLC circuits are known as the second order circuits.

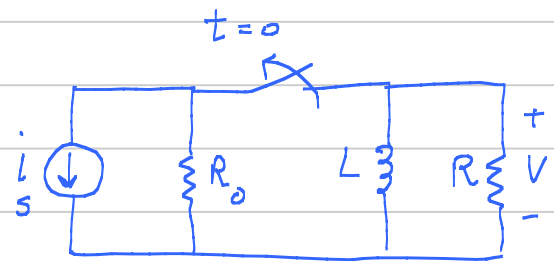
The natural response of an RL circuit

The behavior of a circuit after it is disconnected from a dc source is called natural response.

The switch has been closed for a

long time, meaning all voltages and

currents have reached a constant



value (steady state or dc condition). The inductor behaves as a short circuit

and all the current i_s flows in the inductor branch. When we open the switch

the circuit is reduced to this ckt with the initial inductor current of i_s .

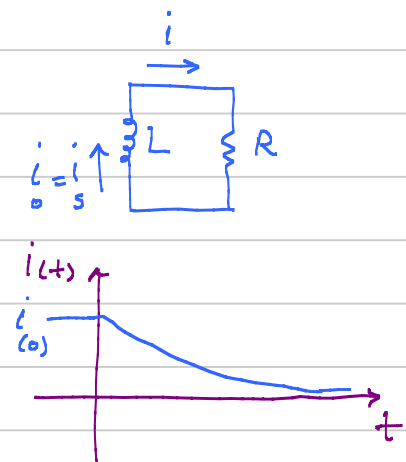
KVL :

$$Ri + L \frac{di}{dt} = 0 \rightarrow L di = -Ri dt$$

$$\frac{di}{i} = -\frac{R}{L} dt \rightarrow \int_{i(t_0)}^{i(t)} \frac{di}{i} = -\frac{R}{L} \int_{t_0}^t dt$$

$$\ln \frac{i(t)}{i(t_0)} = -\frac{R}{L} t \rightarrow i(t) = i(t_0) e^{-\frac{R}{L} t}$$

$$i(0^-) = i(0^+) = i_0 \Rightarrow i(t) = i_0 e^{-\frac{R}{L} t}$$



Time Constant

In the expression for inductor current, the coefficient of t , which is $\frac{R}{L}$ determines the rate at which the current or voltage approaches zero.

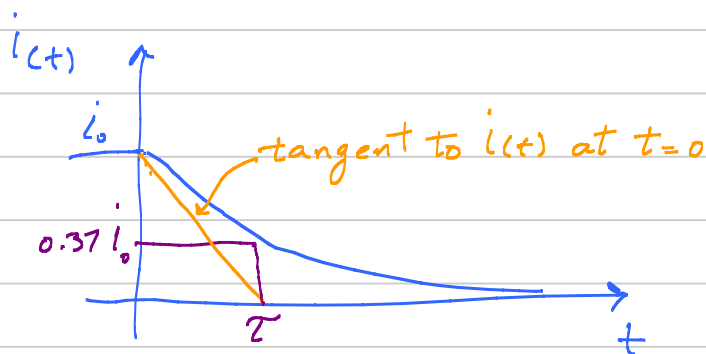
The reciprocal of this ratio is called time constant τ .

$$\tau = \frac{L}{R} \Rightarrow i(t) = \underbrace{i_0}_{\text{initial value}} e^{\underbrace{-\frac{t}{\tau}}_{\text{time constant}}}$$

at one time constant ($t = \tau$), $i(t) = i_0 e^{-1} = 0.37 i_0$

at five time constants ($t = 5\tau$), $i(t) = i_0 e^{-5} = 0.007 i_0 \approx 0$

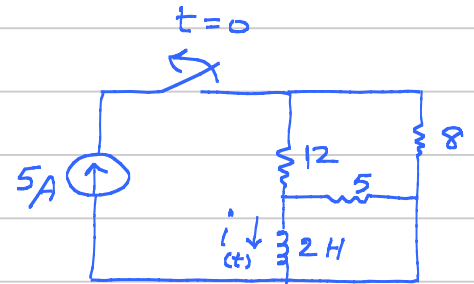
we can assume that after 5 time constants ($t = 5\tau$),
the ckt. has reached the steady state.



Larger $\tau \rightarrow$ slower response

Ex. Find $i(t)$ for $t \geq 0$.

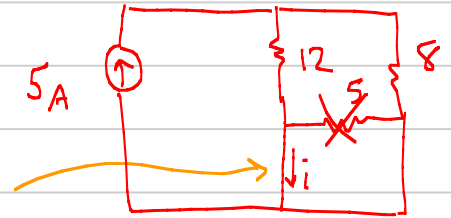
Use the ckt. before switching (at $t=0^-$)
to find the initial value of inductor current.



$$i(0^+) = i(0^-) = i_0$$

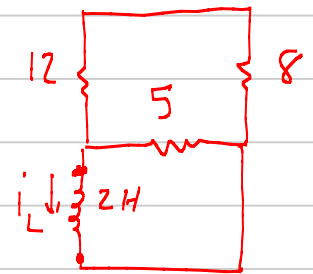
$$i_0 = 5 \frac{8}{12+8} = \underline{2 \text{ A}}$$

inductor is short at steady state ckt. at $t=0^-$



Use the ckt. after switching to find τ .

$\tau = \frac{L}{R_{eq}}$, R_{eq} is the resistance connected
across the inductor's terminal.

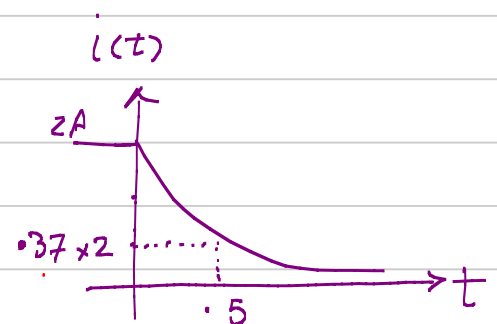


ckt. $t > 0$

$$R_{eq} = 5 \parallel (12+8) = 4 \Omega$$

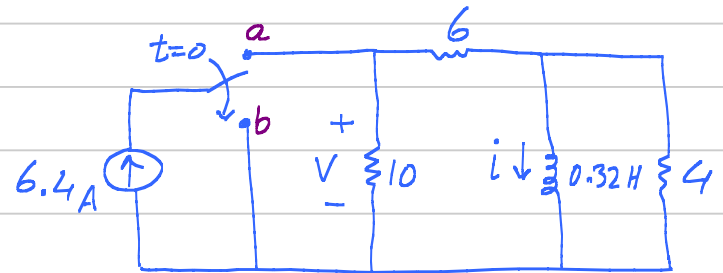
$$\tau = \frac{2}{4} = \underline{0.5 \text{ s}}$$

$$i(t) = i_0 e^{-\frac{t}{\tau}} = 2 e^{-\frac{t}{0.5}} = 2 e^{-2t} \text{ A}$$

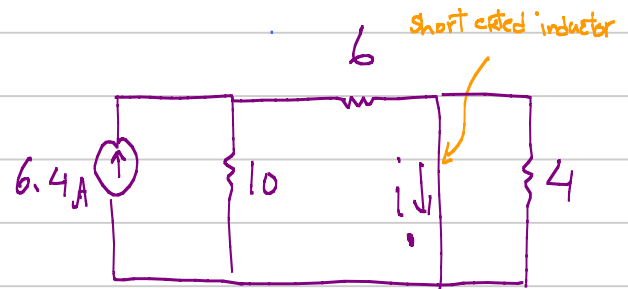


Ex. a) Find v for $t \geq 0$.

b) What percentage of the initial energy stored in the inductor is eventually dissipated in the $4\ \Omega$ resistor?



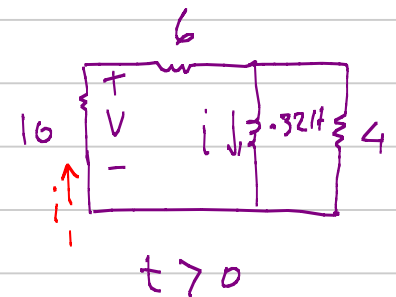
$$i_0 = i_{(0)} = 6.4 \times \frac{10}{10+6} = \boxed{4\text{ A}}$$



ext. at $t=0$

$$R_{eq} = 4 \parallel 16 = 3.2\ \Omega$$

$$\tau = \frac{L}{R_{eq}} = \frac{0.32}{3.2} = \boxed{0.1\text{ s}}$$



$t > 0$

$$i(t) = i_0 e^{-t/\tau} = 4 e^{-\frac{t}{0.1}} = \underline{4 e^{-10t}}$$

CDR:

$$i_1 = \left(4 e^{-10t} \right) \frac{4}{10+6+4} = 0.8 e^{-10t}$$

$$v = -10 i_1 = \underline{-8 e^{-10t}}$$

$$W_{4\Omega} = \int_0^{\infty} P_{4\Omega} dt = 2.05\text{ J}$$

$$\frac{2.05}{2.56} \times 100 = 80\%$$

$$b) W_0 = \frac{1}{2} L i_0^2 = \frac{1}{2} (0.32) (4)^2 = \underline{2.56\text{ J}}$$

$$i_{4\Omega} = i - i_1 = 3.2 e^{-10t} \rightarrow P_{4\Omega} = (4) (3.2 e^{-10t})^2 = 40.96 e^{-20t}\text{ W}$$